

INTERWOVEN EXISTENCE

交織的存在

ECOLOGICAL AI,
ATMOSPHERIC ARCHIVES,
AND CULTURAL STORYTELLING



KINETIC INSTALLATION DRIVEN BY ENVIRONMENTAL WIND DATA FROM LAKE TAIHU, CHINA

HOW CAN TECHNOLOGICAL SYSTEMS
LEARN TO INHABIT THE WORLD WITH US
RATHER THAN PERFORM FOR US?

WHAT IS INTERWOVEN EXISTENCE?

Interwoven Existence is an immersive ecological network that translates a single day of wind conditions from Lake Taihu, China, into continuous physical movement. Visitors by environmental data, encounter climate not as information, but as presence.

The work explores how local environmental conditions shape culture, memory, and everyday life. By combining natural materials with computational systems, the installation brings distant climatic experiences into the gallery, inviting audiences to reflect on the relationships between place, technology, and human experience.

Rather than visualising data, Interwoven Existence performs it. Wind becomes motion, motion becomes atmosphere, and atmosphere becomes a shared space of encounter between people, machines, and the ecological conditions from which both emerge.

WHY DOES IT MATTER FOR DIGITAL HUMANITIES?

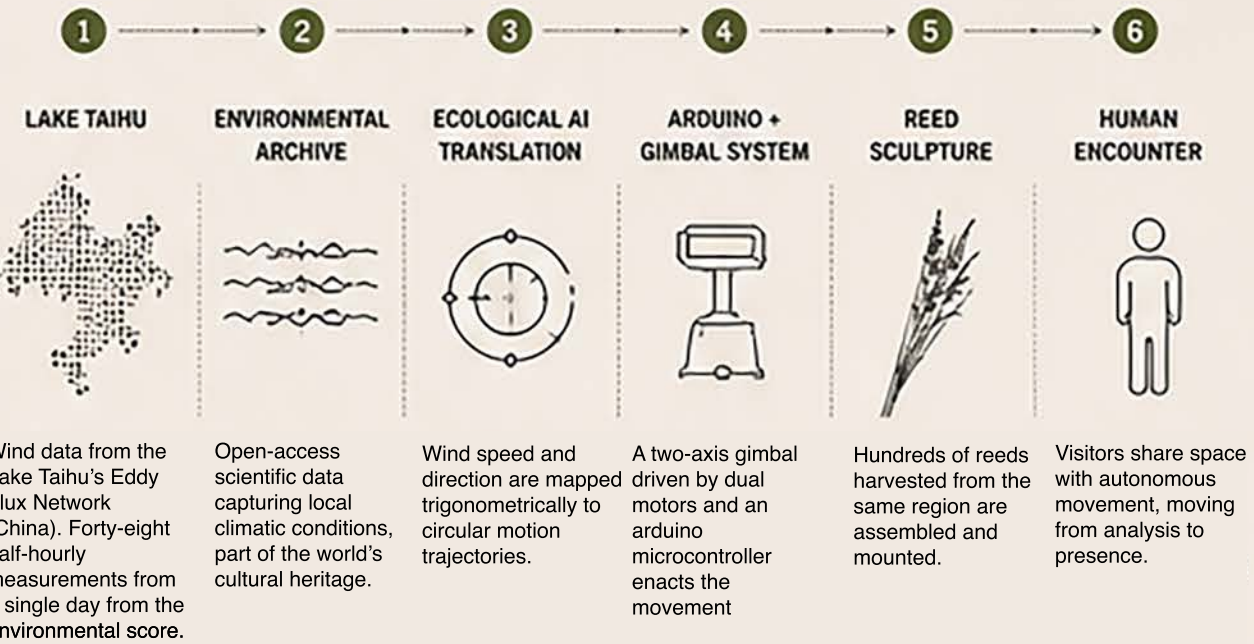
Digital Humanities has transformed access to cultural knowledge, yet much of this dissemination privileges information over experience.

Climate is not merely a backdrop to human activity but an active participant in cultural formation, from poetry and philosophy to economies, migrations, and everyday life.

Interwoven Existence treats environmental data as an atmospheric archive, a record of lived ecological conditions. By translating wind into embodied movement, the work extends DH beyond representation towards experiential encounter.

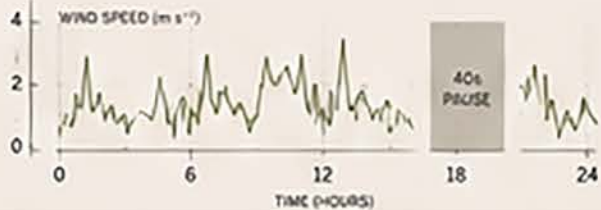
It invites us to preserve and communicate not only what happened in a place, but what it felt like to be there.

HOW IS IT CREATED?



DESIGNED AUTONOMY

A 40-second pause interrupts movement between sequences. This intentional delay breaks the perception of reactivity, inviting audiences to experience the work as an autonomous presence, like wind, reeds, or trees, unfolding in its own time.



WHAT DOES THIS PROJECT SUGGEST ABOUT AI?



Most contemporary AI systems are trained on human-generated data and optimised to serve human goals.

Ecological AI looks beyond the human, drawing on non-human data and relational intelligence.

Interwoven Existence is not adaptive, predictive, or responsive to audiences. It operates according to an environmental score and exists alongside us.

The project asks whether intelligent systems might cultivate coexistence rather than performance.

It shifts the question from "How can machines become more like us?" to "How can technological systems learn to inhabit the world with us?"

天人合一

The philosophy of "unity of heaven and humanity" inspires a vision of coexisting with natural processes.

CONTRIBUTIONS TO DIGITAL HUMANITIES



Environmental experience as cultural heritage
Proposes atmospheric memory and historical interpretation.



Performative storytelling
Demonstrates how environmental datasets can be embodied to tell stories that screens cannot.



Ecological AI for cultural technologies
Offers an alternative framework to dominant paradigms of optimisation, personalisation, and human imitation.



Attentiveness and coexistence
Encourages contemplative practices and new modes of engagement with cultural and ecological systems



DATA SOURCE

Lake Taihu Eddy Flux Network
Zhang et al. (2020)
Earth System Science Data
DOI: 10.5194/ESSD, 12, 2635–2645, 2020

Half-hourly measurements of wind, radiation, temperature, humidity, and energy fluxes collected at multiple sites across Lake Taihu, China from 2010–2018.



● Eddy covariance sites

MATERIAL / PROCESS



REEDS HARVESTED LOCALLY



ASSEMBLED INTO BUNDLES



MOUNTED ON 2-AXIS GIMBAL



ARDUINO + DUAL MOTORS



INSTALLED AS LIVING ARCHIVE

EXHIBITIONS

Hanshan Museum
Suzhou, China

Ars Electronica 2024
Linz, Austria

AUTHORS

Dr Andrew O'Dowd
Dr Isadora Petruskas
Mr Zhonghao Chen



Wind has no shape, yet leaves traces. Data has no body, yet carries presence. When translated through matter, we may share a moment, not as masters, but as participants in the world.



INFLUENCES

Johanna Drucker - CAPTA
Katherine N. Hayles - How We Became Posthuman
Yuk Hui - On the Existence of Digital Objects and Cosmotechnics

